

MEDICAL PHYSICS

Lab. Electrocardiogram (ECG)

A.L. Ruaa Emad

Supervised by

Assist. Prof. Dr. Shreen Hajy

ruaa.hussien@hmu.edu.krd

Medical Physics Unit

2nd semester

1st stage

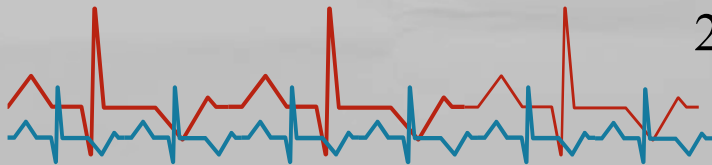
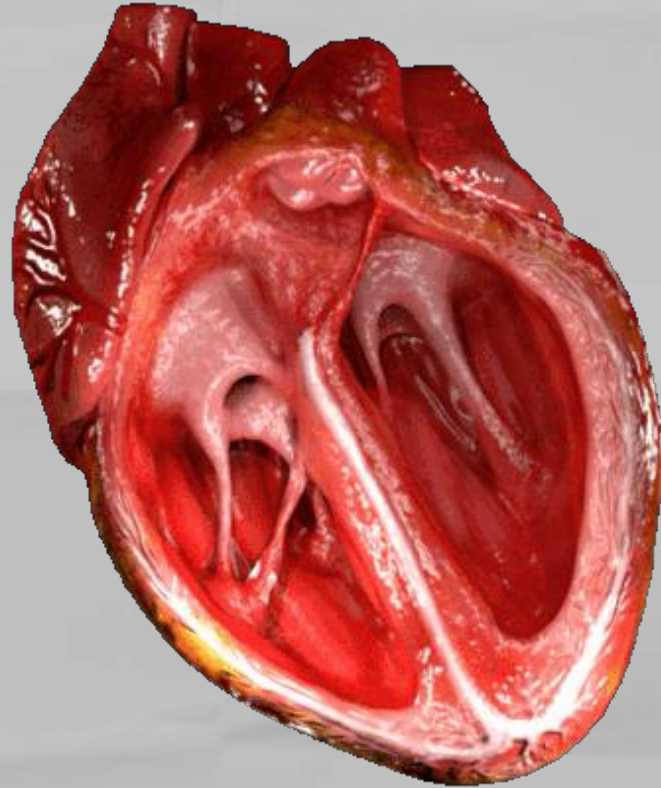


TABLE OF CONTENTS

01

Anatomy of the
Cardiovascular
System

02

Cardiac Conduction
System

03

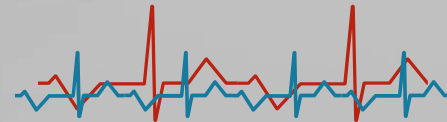
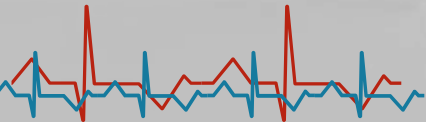
ECG

04

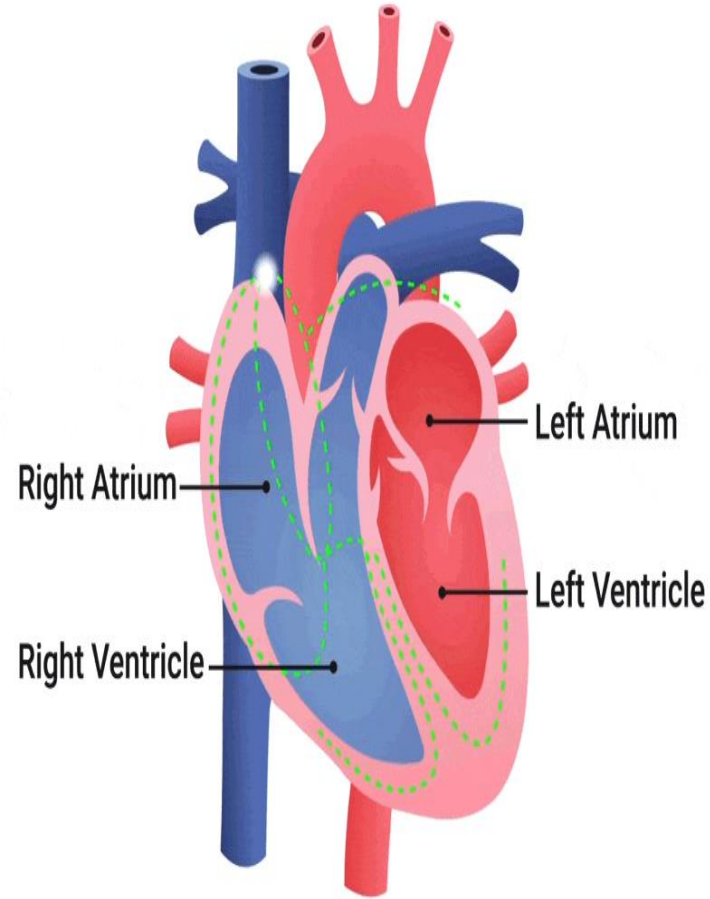
ECG Paper

05

Patient preparation



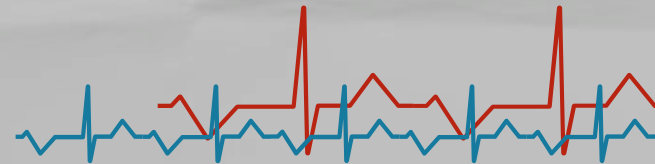
Cardiovascular System

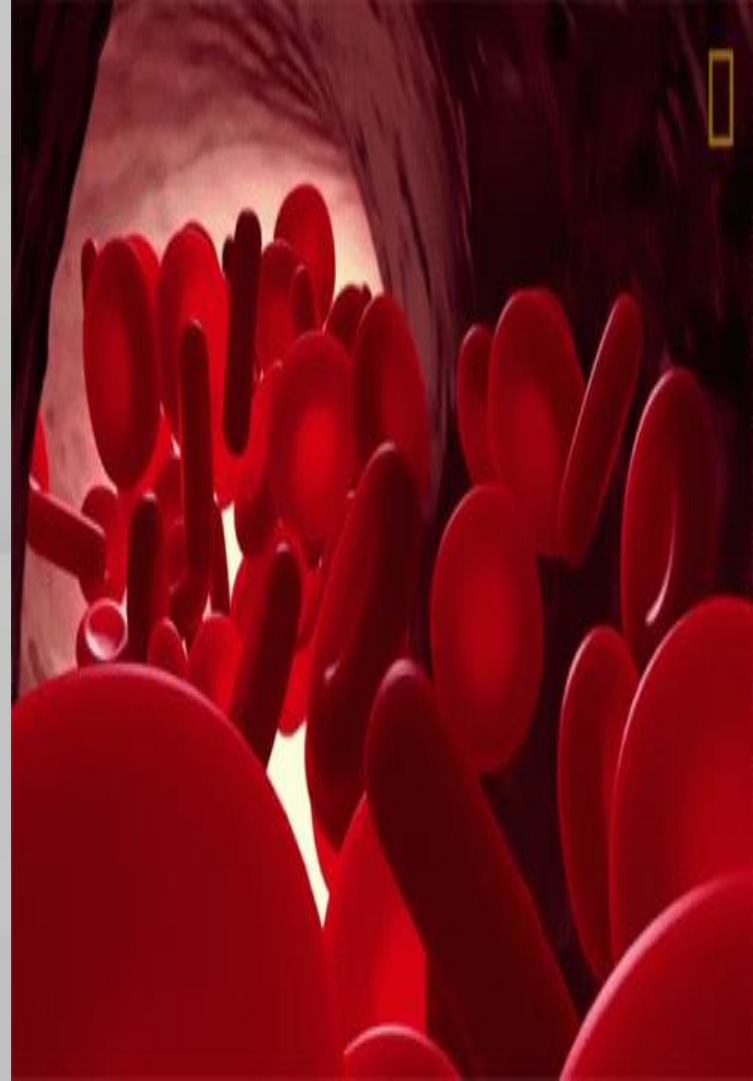


1. Heart:

Structure: Four chambers - two atria and two ventricles.

Function: Acts as a pump to circulate blood throughout the body.



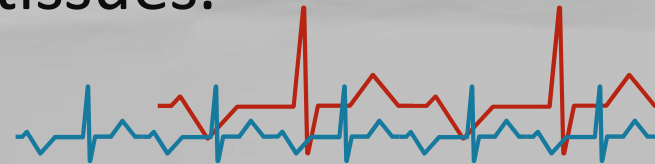


Anatomy of the Cardiovascular System

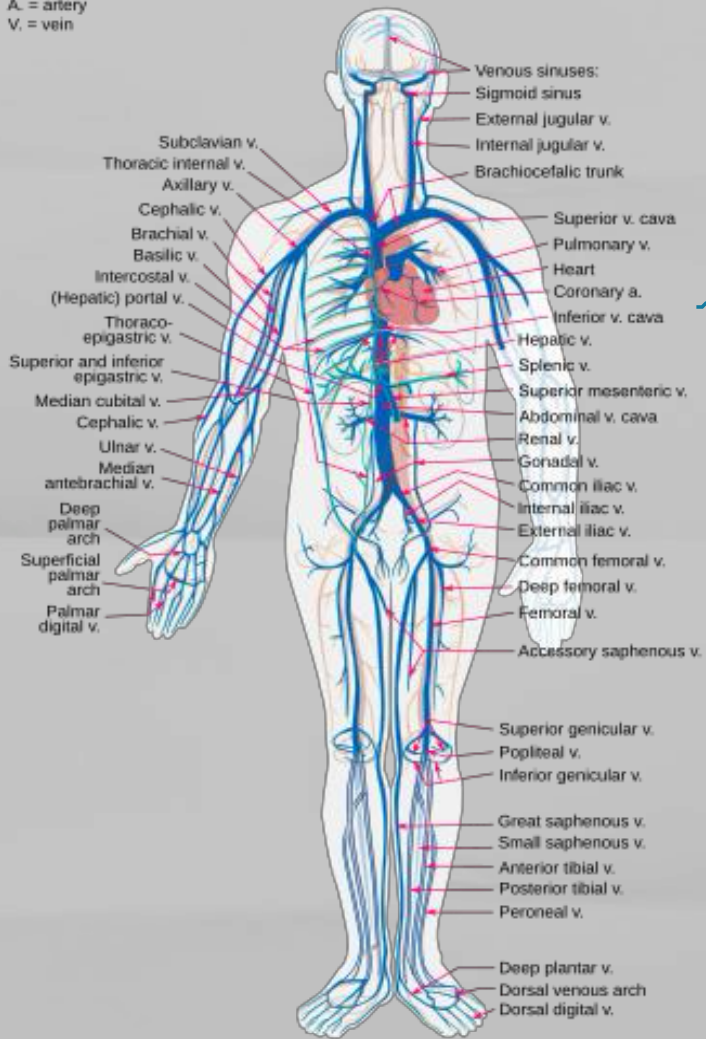
2. Arteries:

Structure: Thick-walled vessels carrying oxygenated blood away from the heart.

Function: Distribute oxygen and nutrients to tissues.



A. = artery
V. = vein

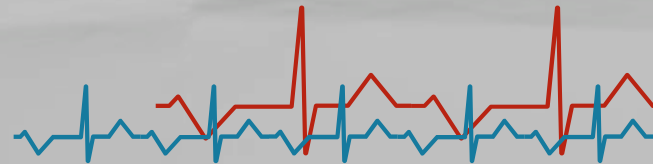


Anatomy of the Cardiovascular System

3. Veins:

Structure: Thin-walled vessels carrying deoxygenated blood towards the heart.

Function: Return blood to the heart.

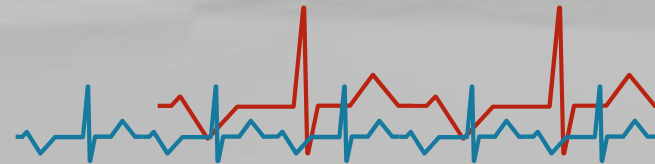
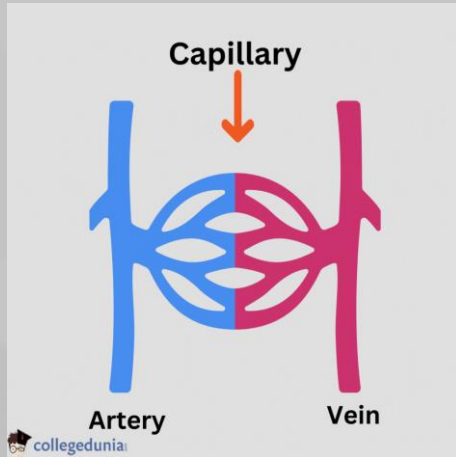


Anatomy of the Cardiovascular System



4. Capillaries:

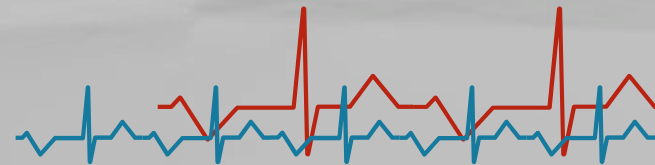
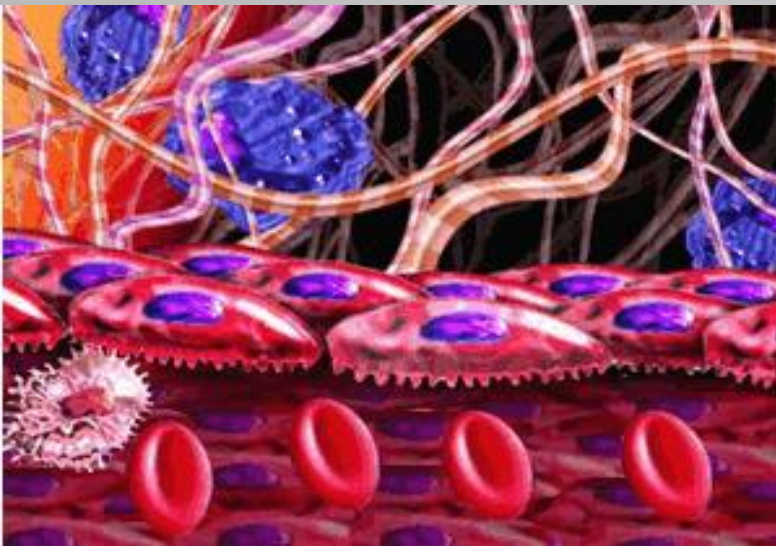
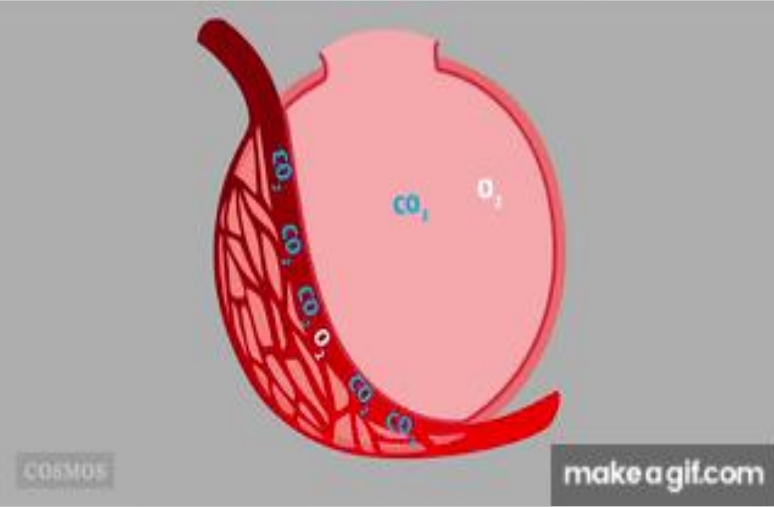
Structure: Microscopic vessels connecting arteries and veins.



Anatomy of the Cardiovascular System

4. Capillaries:

Function: Facilitate exchange of oxygen, nutrients, and waste products.

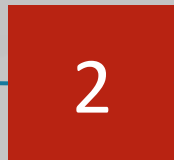


Cardiac Conduction System

Sinoatrial
node

Bundle of His

SA node



Atrioventricular
node

Purkinje
fibers

AV node

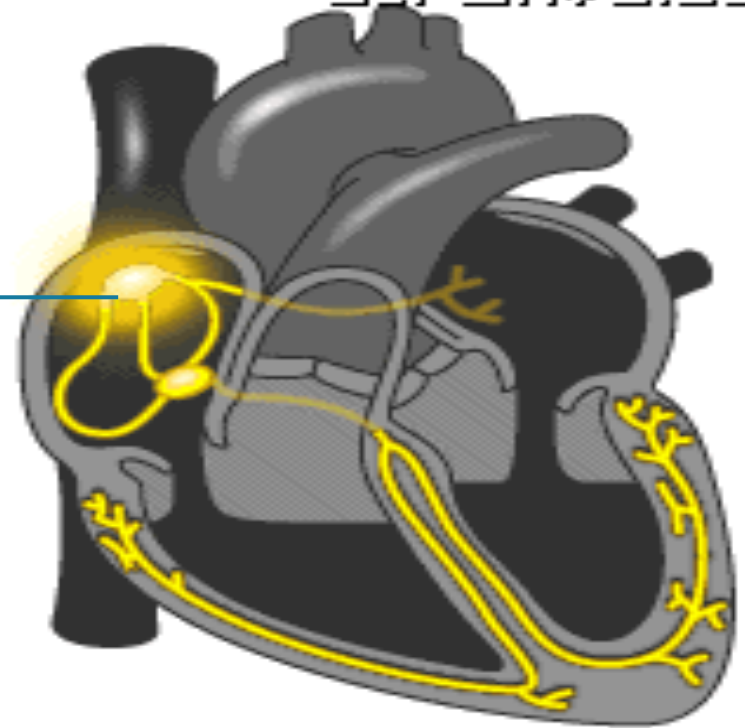


Cardiac Conduction System

Sinoatrial
node

SA node

1



Cardiac Conduction System

STFWAVE.COM

Sinoatrial
node

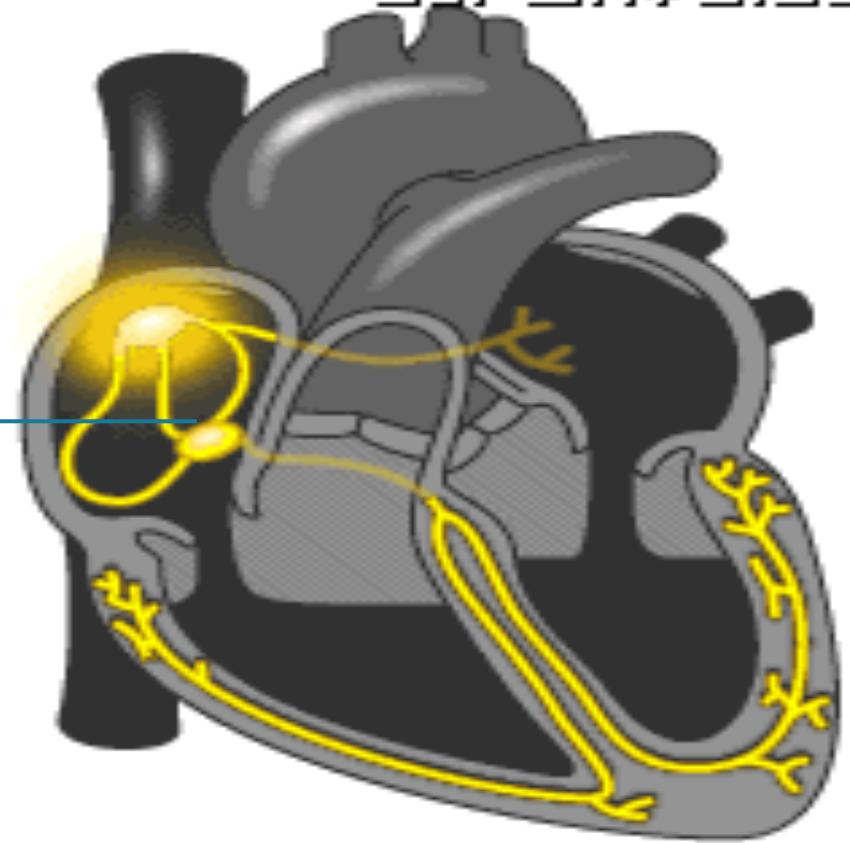
SA node

1

2

Atrioventricular
node

AV node



Cardiac Conduction System

Sinoatrial
node

SA node

1

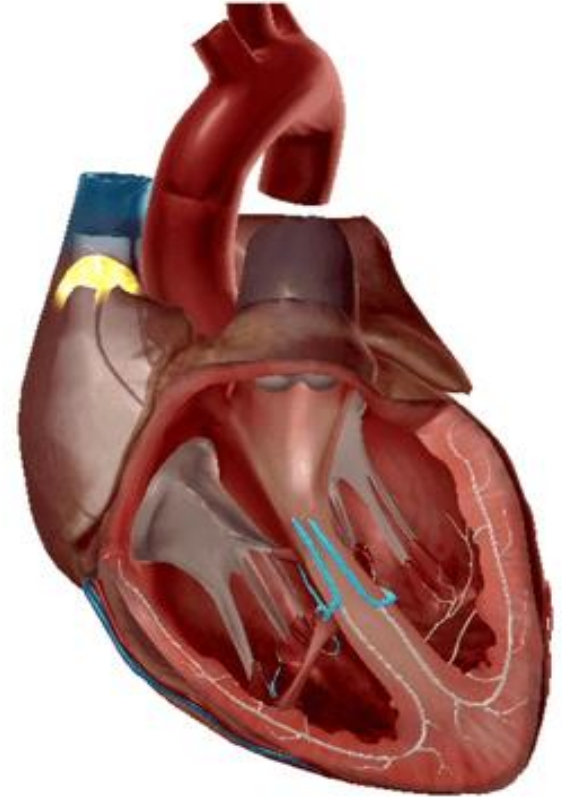
2

3

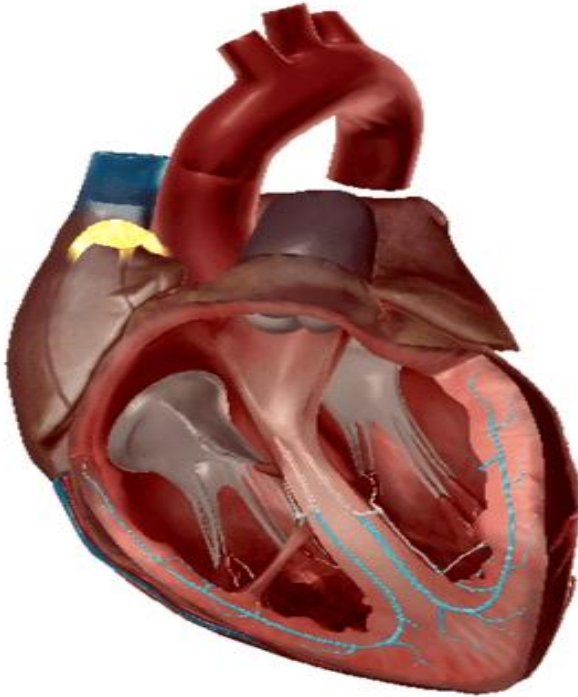
Bundle of His

Atrioventricular
node

AV node

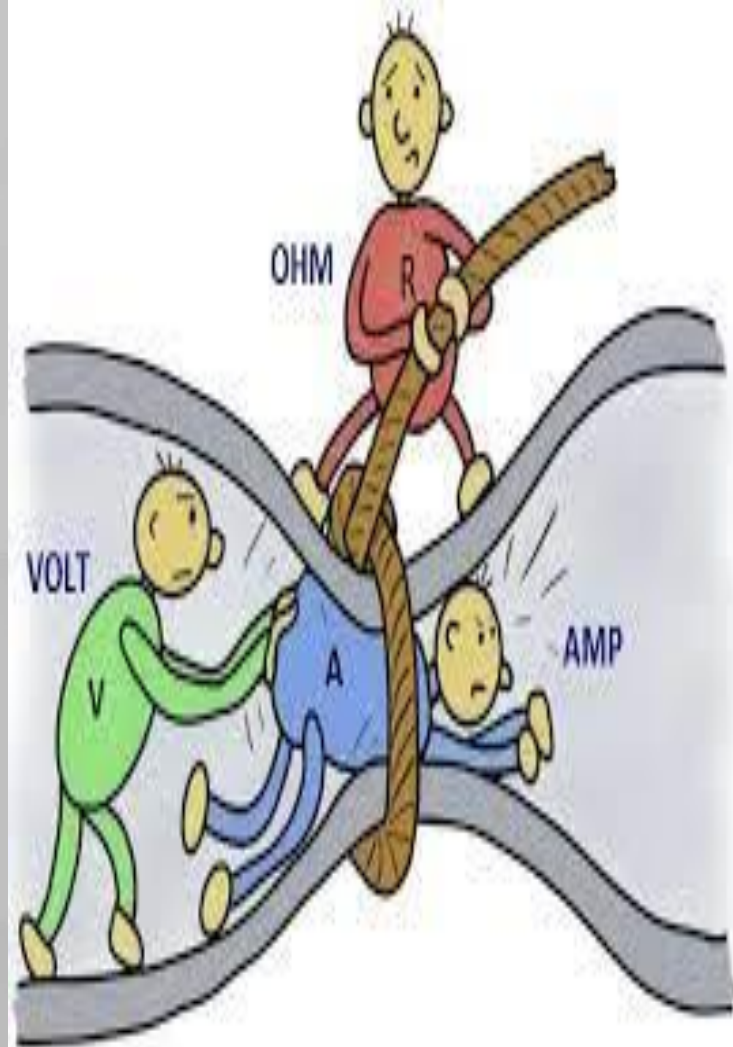


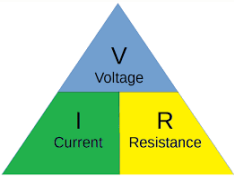
Cardiac Conduction System



Ohm's Law:

Ohm's Law is applied to understand the relationship between voltage, current, and resistance in the electrical circuits of ECG devices.





Ohm's Law states that the current flowing through a conductor between two points is directly proportional to the voltage across the two points and inversely proportional to the resistance between them.



Is a diagnostic tool used to measure and record the electrical activity of the heart over a period of time.

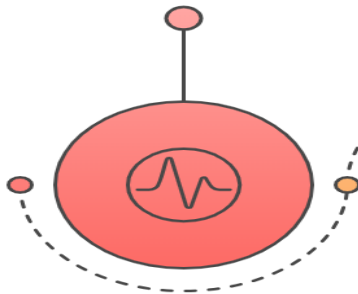
How does (ECG)Work:



Electrical Signal Flow in ECG

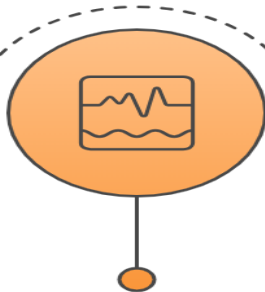
Heart Generates Signals

The heart produces electrical signals.



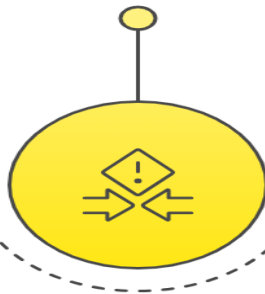
Signals Flow Through Tissues

Signals travel through body tissues.



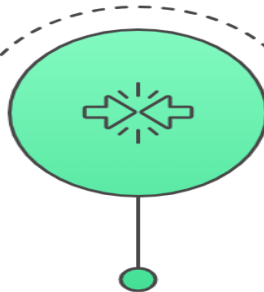
Signals Encounter Resistance

Skin and tissues resist signal flow.



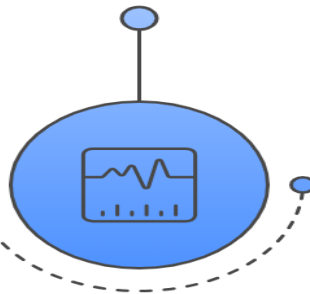
Current Reaches Electrodes

The current reaches the ECG electrodes.



Voltage Recorded by ECG

The ECG records the potential difference.



Components of ECG



Leads



Wires



Amplifier



Display



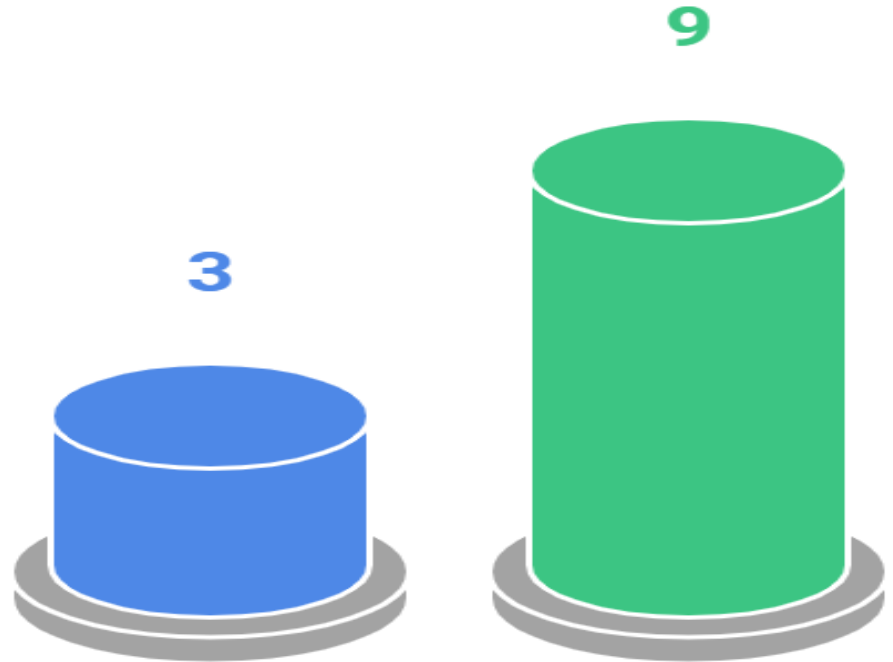
Distribution of ECG Leads

Bipolar Leads

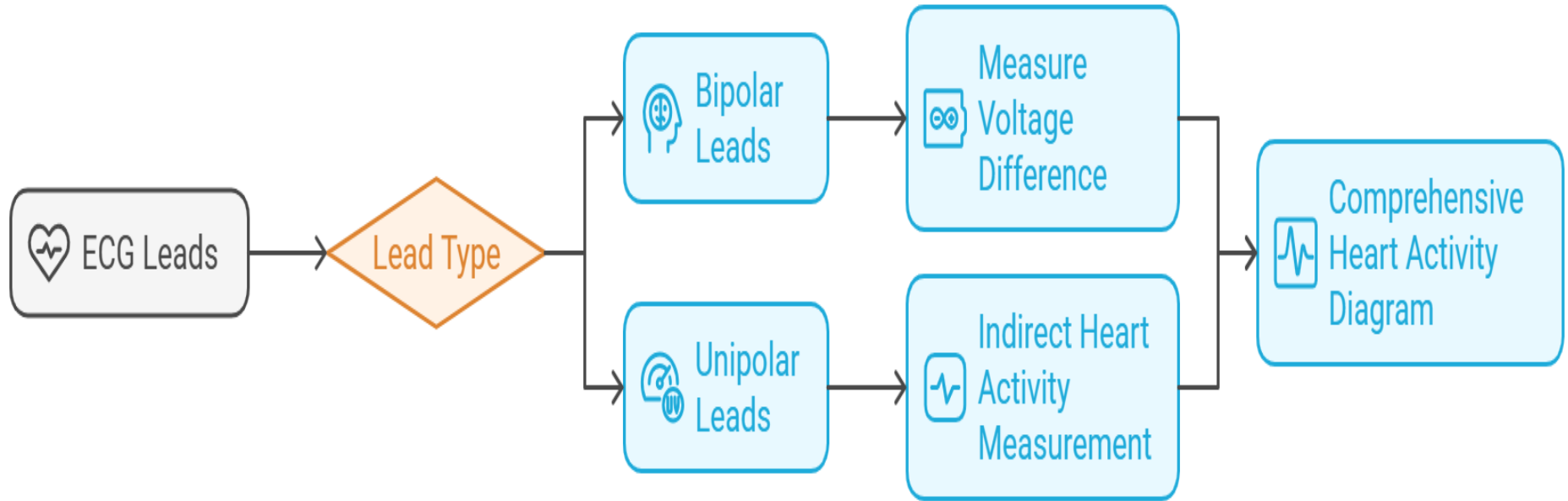
Measure voltage difference either between the arms or between an arm and legs

Unipolar Leads

Measure voltage from a single point



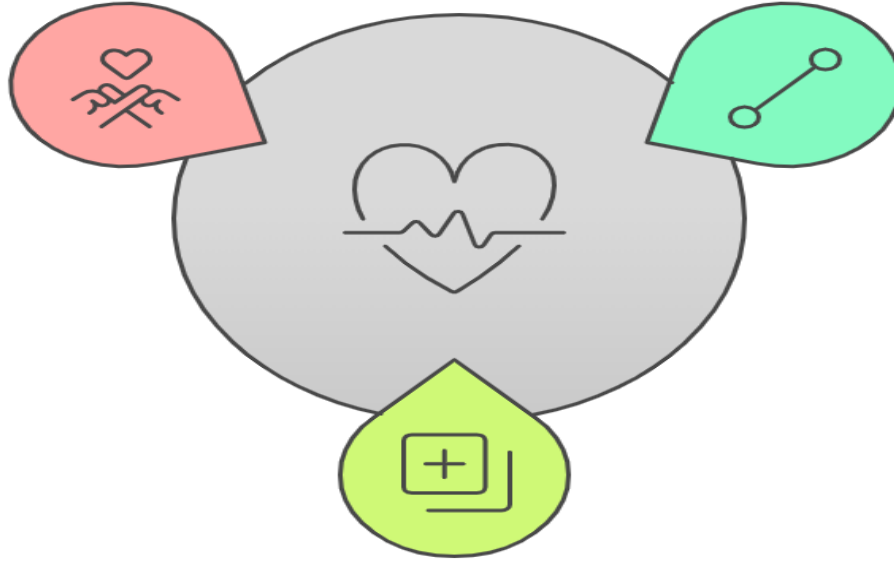
ECG Leads and Heart Activity



Components of ECG Leads

Chest Leads

Provides detailed heart activity from the chest area



Bipolar Limb Leads

Measures electrical activity from two points on the body

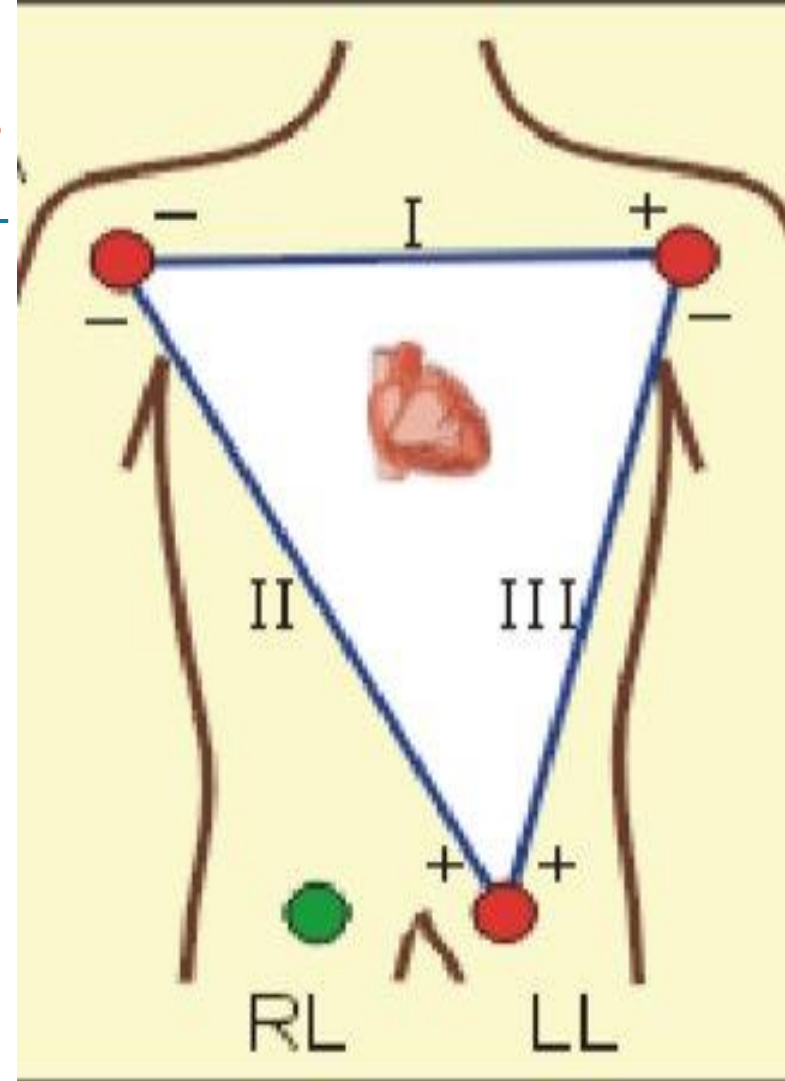
Augmented Leads

Enhances signal detection by using a single positive electrode

Standard limb leads (bipolar).

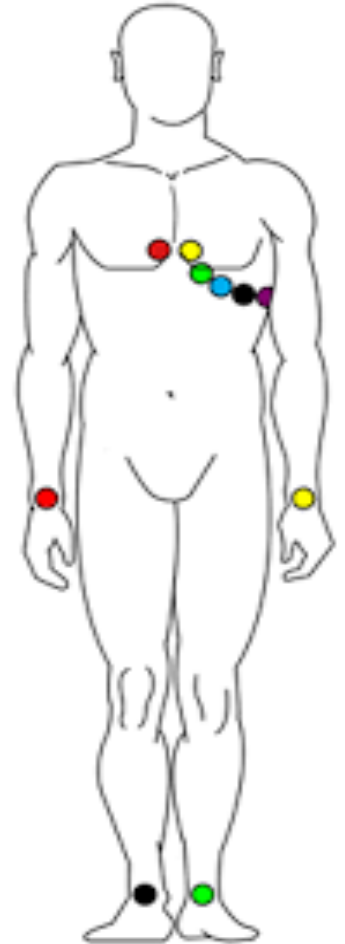


- **Lead I** = right arm(RA) –ve, left arm(LA) + ve.
- **Lead II** = right arm (RA) _ ve ,left leg (LL)+ ve.
- **Lead III** = left arm(LA) _ ve, left leg(LL) +ve.



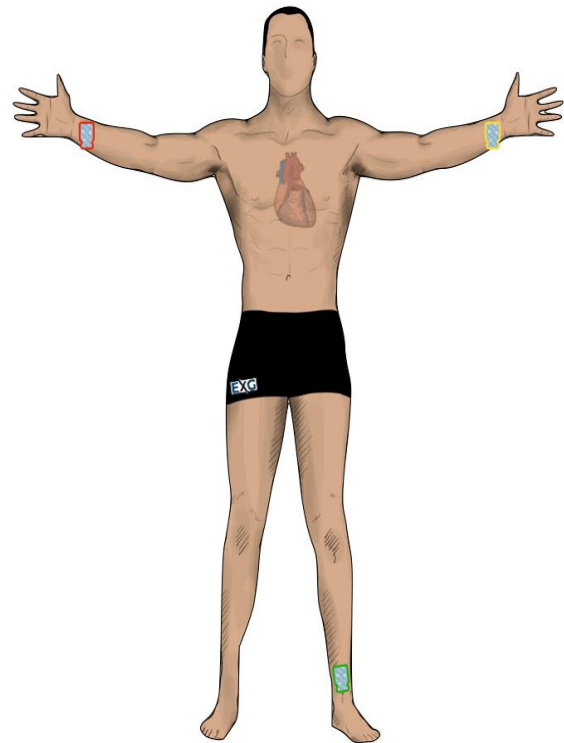
Standard Limbs leads

- The limb leads are color-coded to avoid misplacement (red - right arm, yellow - left arm, green - left leg, and black - right leg).



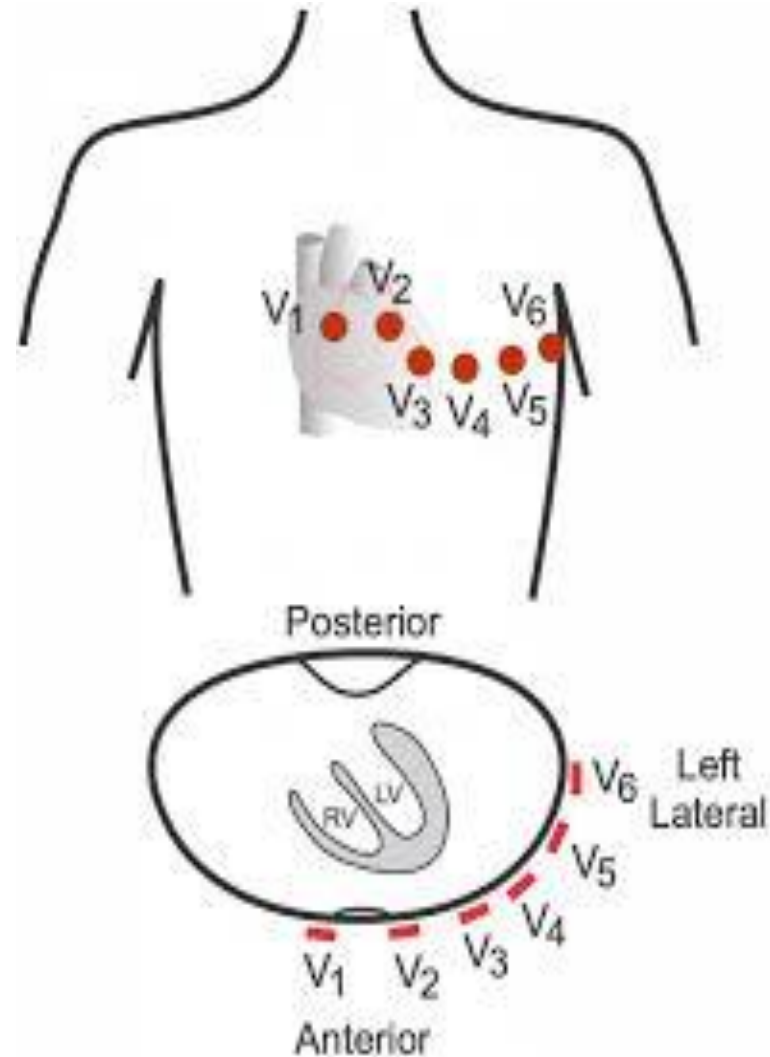
Augmented limb leads(unipolar)

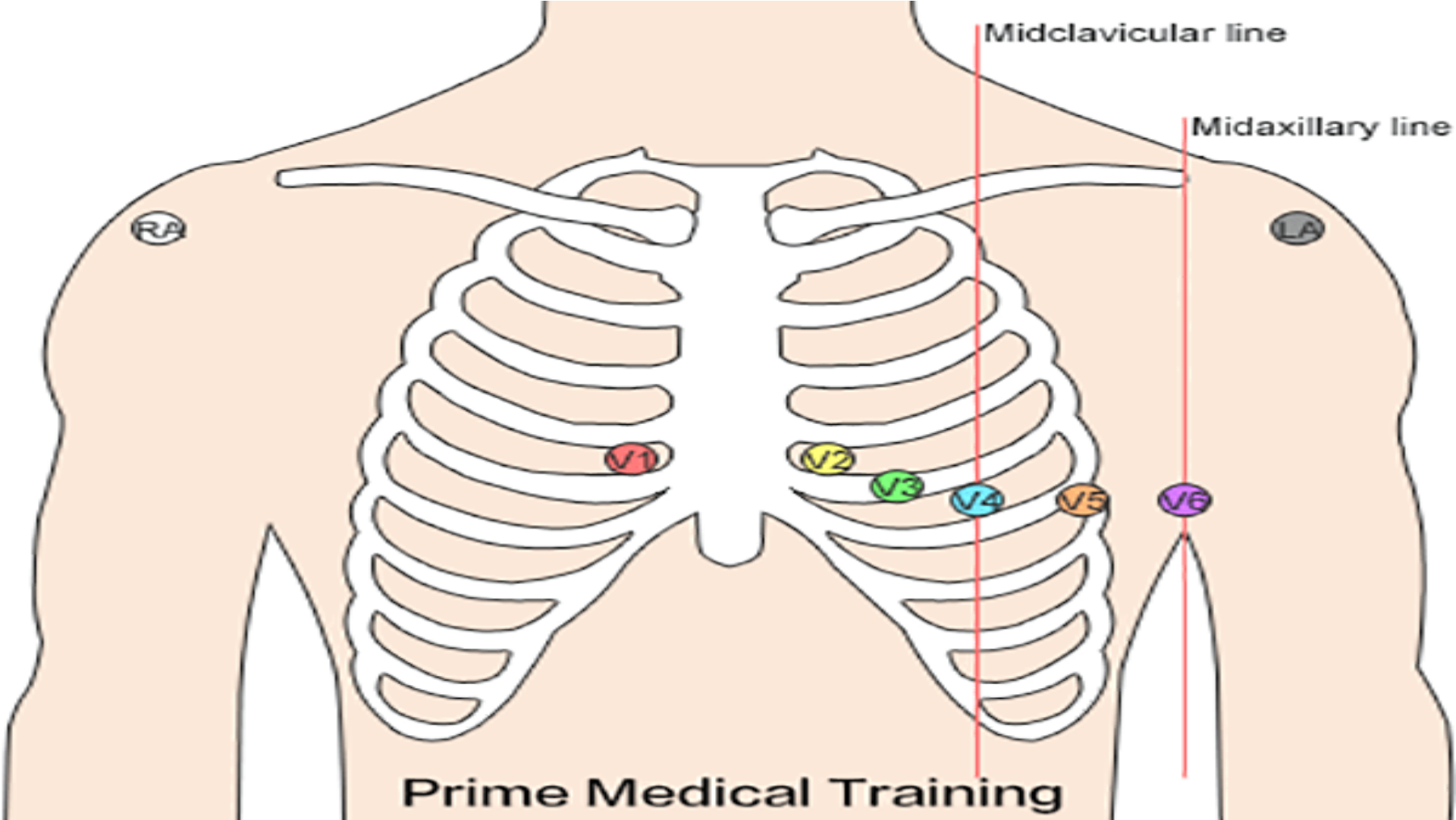
- aV_R attach to right arm.
- aV_L attach to left arm.
- aV_F attach to left foot.



Precordial leads (Unipolar)

The precordial leads V1 to V6 are attached to the surface of the chest.





Midclavicular line

Midaxillary line

RA

LA

V1

V2

V3

V4

V5

V6

Prime Medical Training

How ECG works?



Leads

Electrodes that detect and record the electrical signals produced by the heart.



Wires

Channels through which the electrical signals are transmitted from the body to the amplifier.



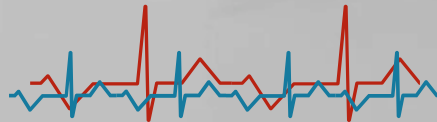
Amplifier

Device that strengthens the weak electrical signals detected by the electrodes for better visualization.



Display

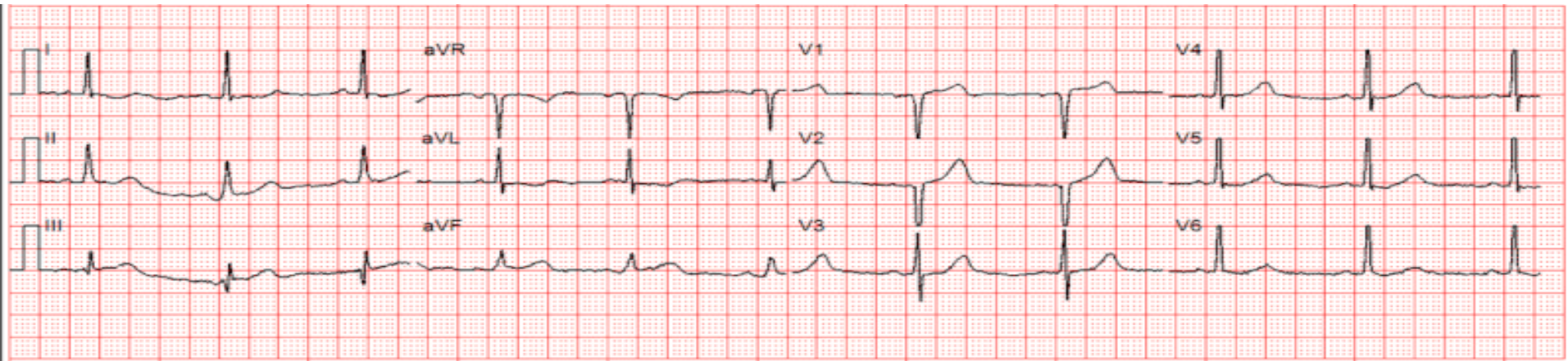
Interface where the amplified signals are visually represented for analysis and interpretation.



Electrocardiogram paper

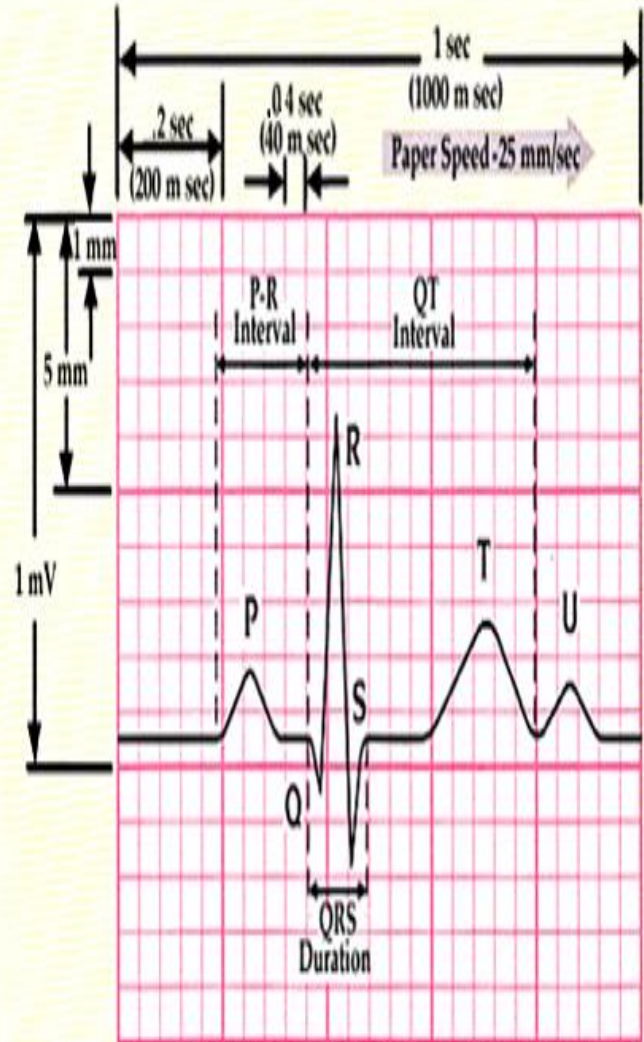


Is the graph on which this electrical activity is recorded.



paper

- This is long role of paper, composed of small squares.
- One square is 1mm wide and 1 mm high.
- On ECG paper there are thick lines, between two thick lines there are 5 small squares.
- The speed of ECG machine is 25 mm per second.



ECG Paper

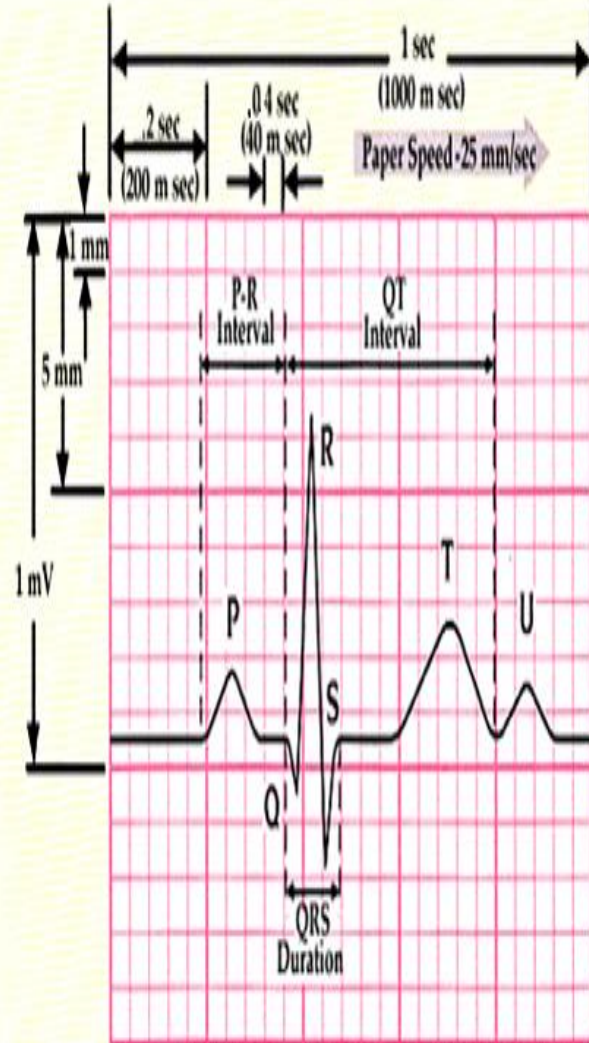


- The time interval of each small square can be calculated as:
- 25 small sqrs are equal to 1 second.
- 1 small sqr is equal to $1/25$ second.
- i.e 0.04 seconds.

ECG Paper

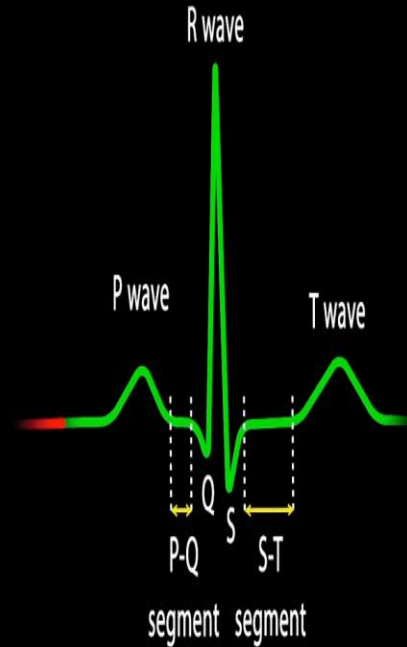
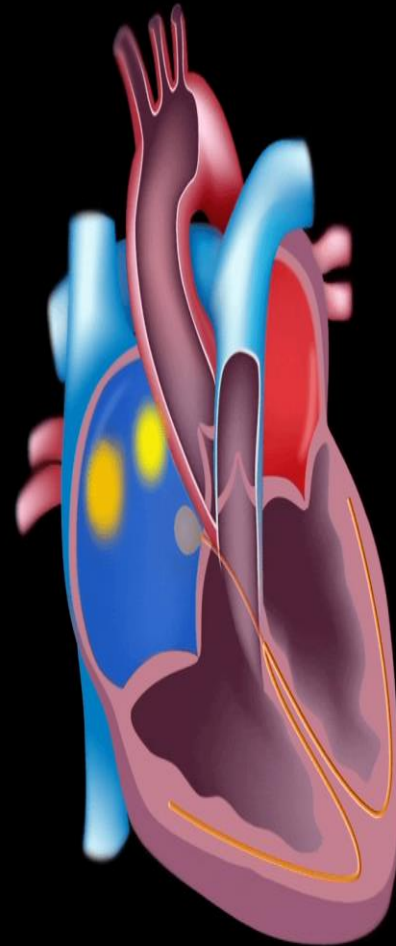


- vertically the small square represent the amount of electrical potential.
- One small square represent the potential of 0.1 mV.
- 10 small squares represent the potential of 1 mV.



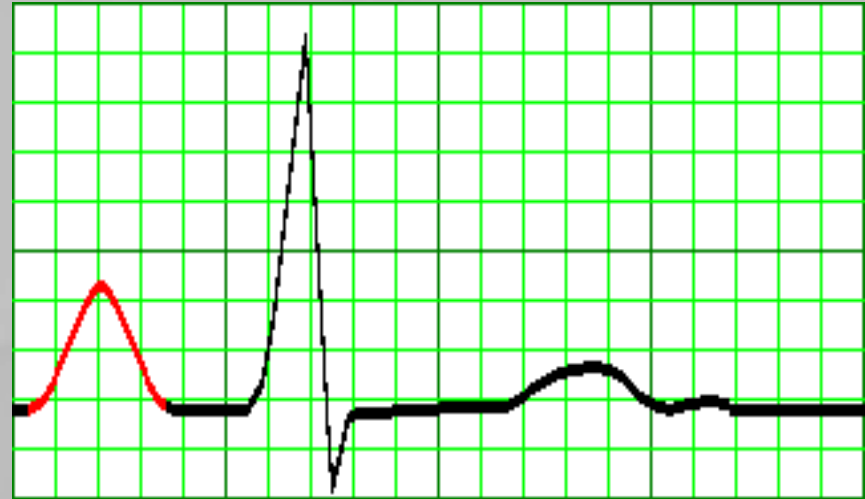
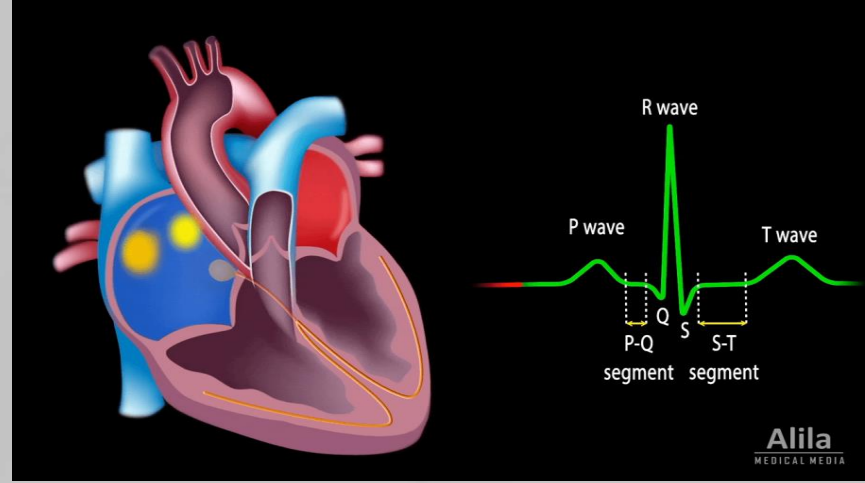
Wave forms

- P wave: Atrial depolarization
- QRS complex: Ventricular depolarization
- T wave: Ventricular repolarization



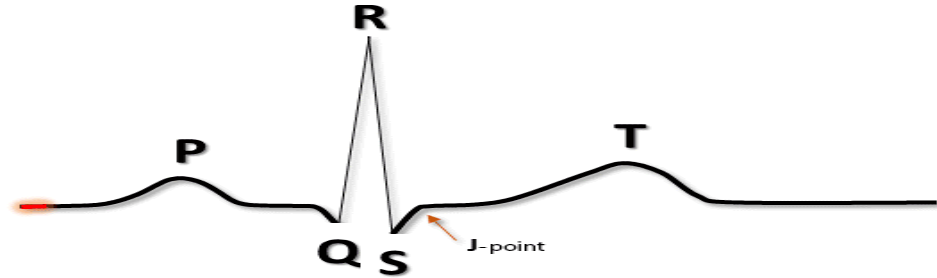
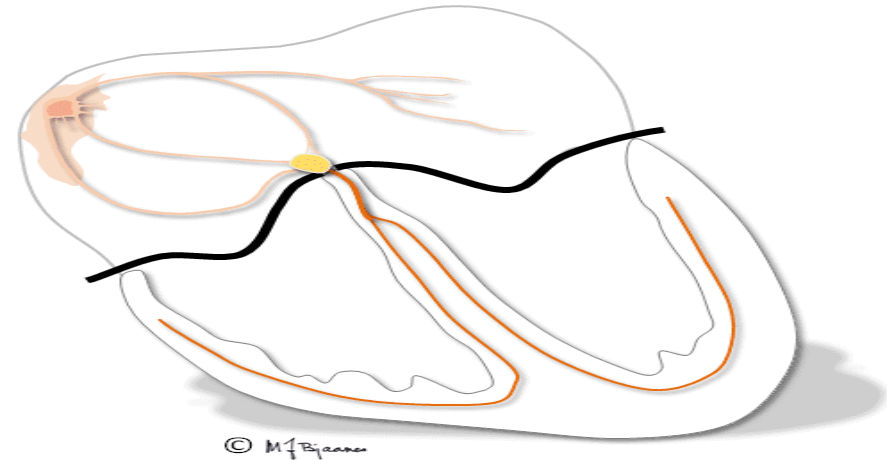
P wave

- P Wave shows atrial depolarization.
- Its duration is 0.1 sec (2 and half small sqr) and height is 2.5 mv (2 and half small sqr).



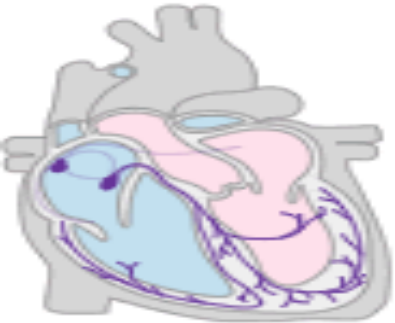
QRS complex.

- QRS complex represent the ventricular depolarization.
- its normal duration is about 0.08-0.1 S (less than 2 small sqrs) and height is about 5 to 20 small sqrs.

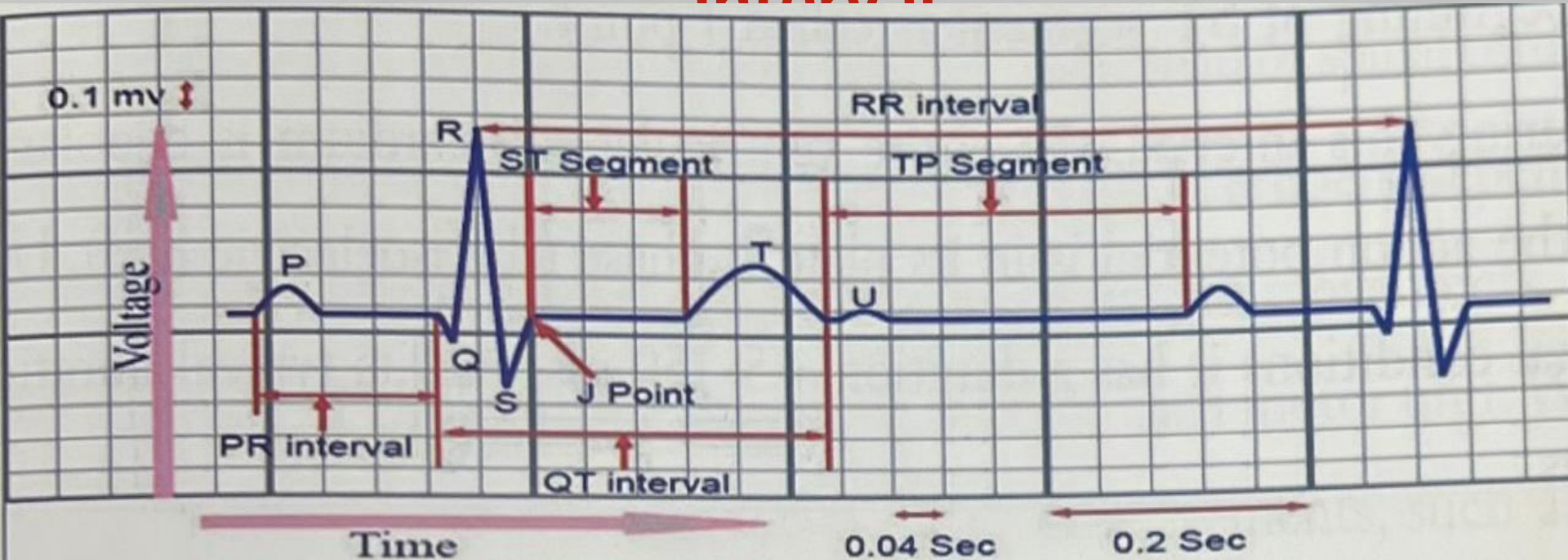


T wave

- P Wave shows ventricular repolarization.
- Its duration is 0.16 sec.



Normal ECG interval

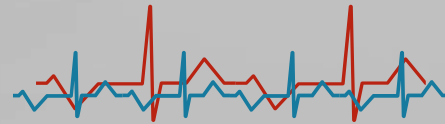


- PR interval 0.12 – 0.20 sec
- QRS duration 0.08 – 0.10 sec

- QT interval 0.4 – 0.43 sec
- RR interval 0.6 – 1.0 sec

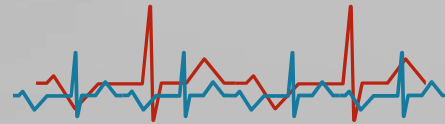
ECG helps to know?

- Heart Rate
- Rhythm
- Wave morphology
- Intervals and segments analysis
- Chamber enlargement
- Specific changes



How to calculate heart rate.

- **Rule of 300**
- Take the number of "big boxes" between neighboring QRS complexes, and divide this into 300. The result will be approximately equal to the rate.



The Rule of 300

It may be easiest to memorize the following table:

# of big boxes	Rate
1	300
2	150
3	100
4	75
5	60
6	50

Heart rate ?



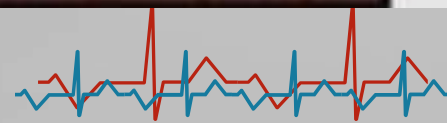
Heart rate ?

$HR = 300 / 4 = 75 \text{ bpm}$

Heart rhythm: Sinus normal rhythm



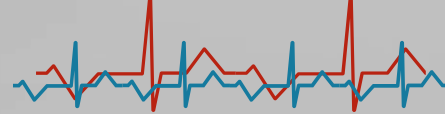
Heart rate ?



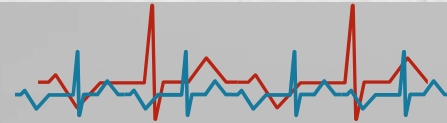
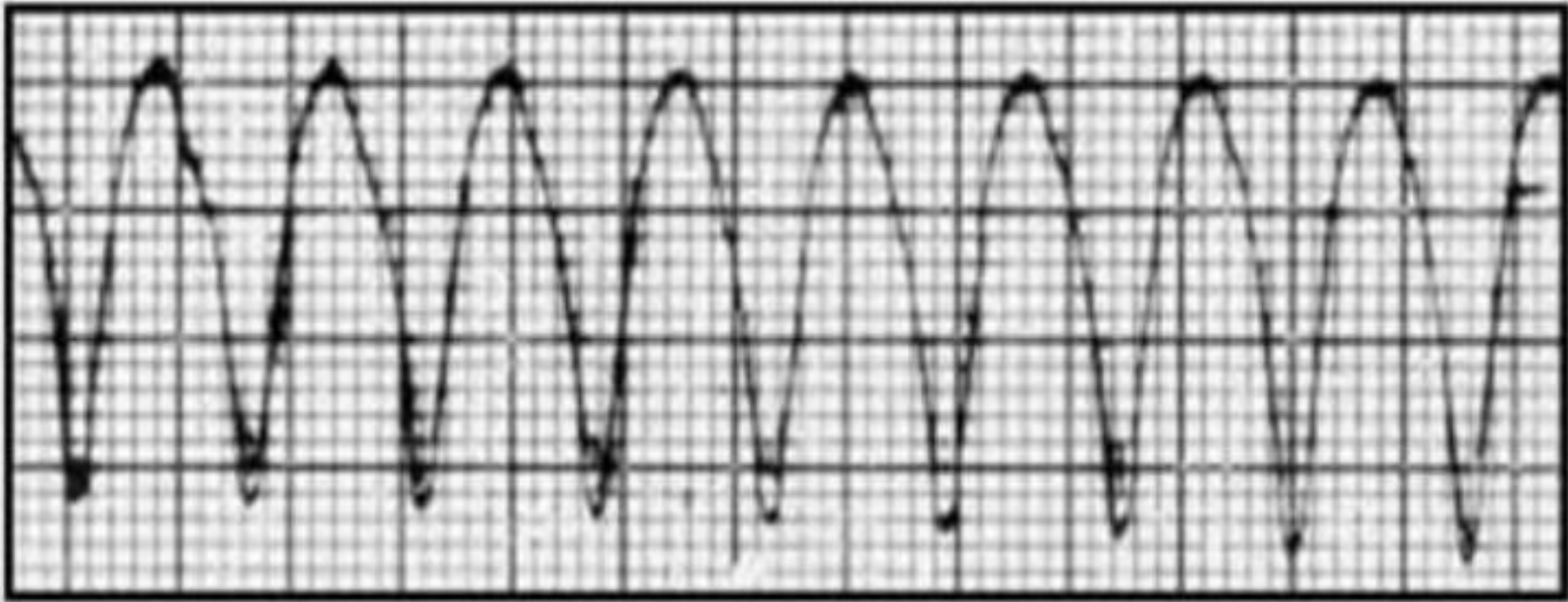
Heart rate ?

$$\text{HR} = 300 / 6 = 50 \text{ bpm}$$

Heart rhythm: Sinus bradycardia



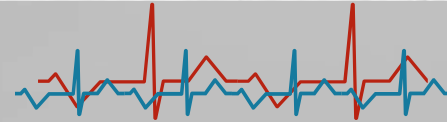
Heart rate ?



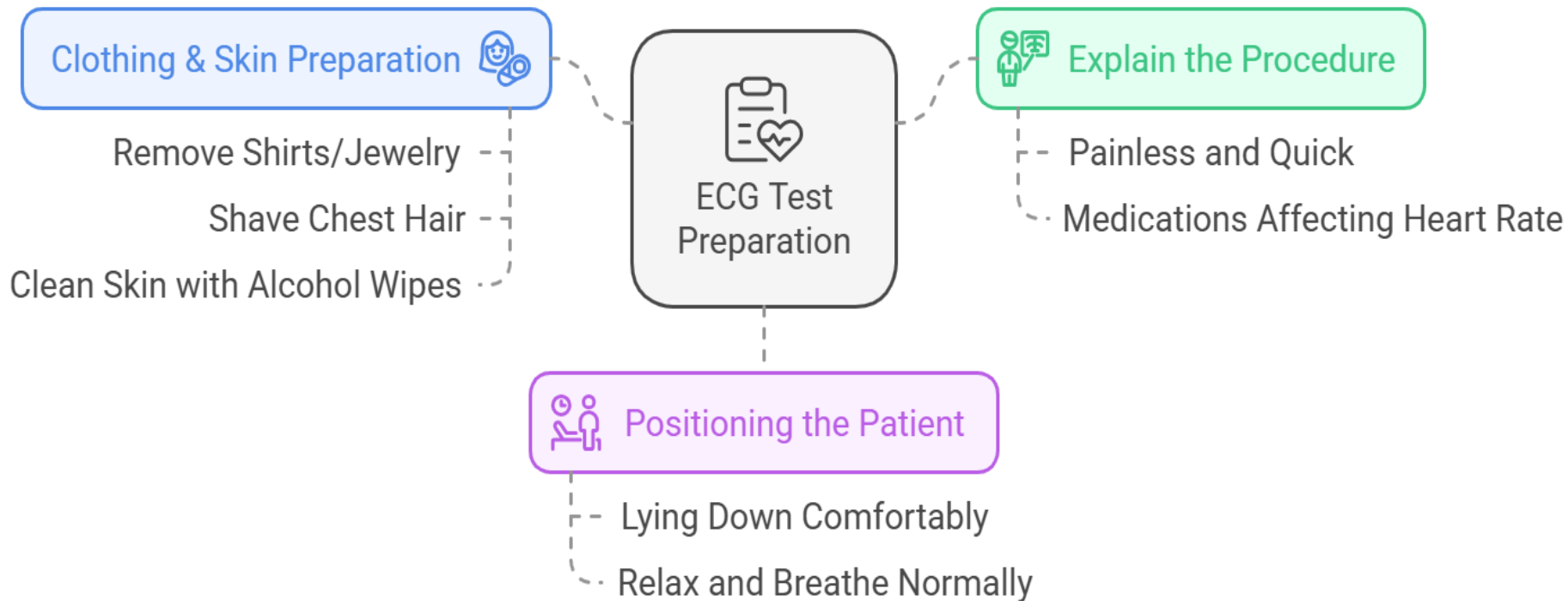
Heart rate ?

HR= $300/1.5 = 200\text{bpm}$

Heart rhythm: tachycardia



ECG Test Preparation: Key Steps and Considerations



ECG Test Procedures



Electrode Placement

Place electrodes on the body for ECG.



Avoid Movement

Instruct patient to remain still during test.

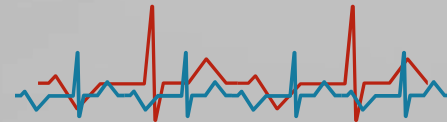
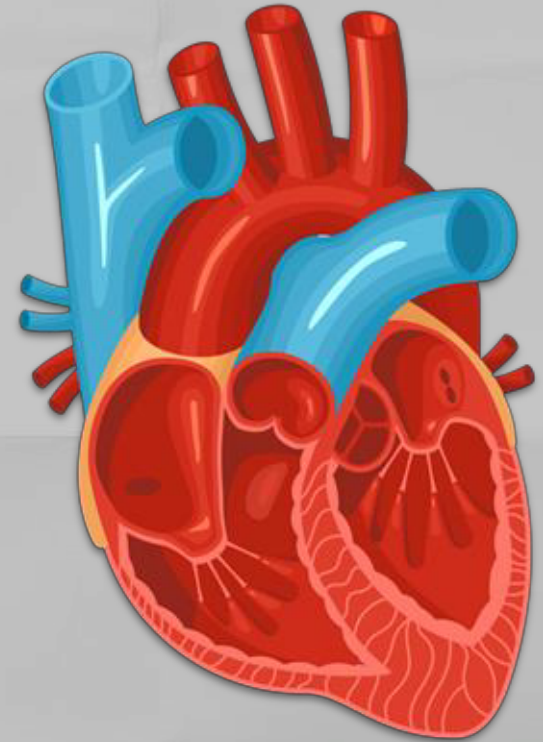


Check Connections

Verify all lead connections are secure.

REFERENCES

- PROF DR TAKI A M AL MUSAWI (2024). MEDICAL PHYSICS FOR MEDICAL STUDENTS
- JOHN R CAMERON . MEDICAL PHYSICS



THANKS

